Week 7

Prof. Michael Sandel of philosophy argued that the hypothetical question in philosophy, the trolley dilemma has now become the real problem in programming driverless cars.
From an engineering point of view, it might be possible to apply some philosophical principles to the program once they are given. However, the largest hurdle to handling this problem set is the UNCERTAINTY of the information the program should rely on just as I showed you in the last part of today's class, predicting how bicyclists or pedestrians behave largely depends on the detailed information and the contexts, both of which are missing in the state of the art autonomous cars.

Suppose you are a director of software engineers. Think of a specific situation in an autonomous vehicle where there is UNCERTAINTY about information that could change the moral decision depending on the situation. We want a program that can act like a human being in the face of this uncertainty, so describe in 400 words how you would behave in such an uncertain situation so that the programmers can implement it. The moral principle can be assumed in any way and should be clearly stated.

Grading criteria: Clearness, concreteness, reality, and originality (not easily found on the Net, should not be same as presented in today's class)

Answer
Consider the following scenario: An autonomous vehicle is cruising down a busy road when a group of children unexpectedly darts into the street from behind a parked car. The vehicle's sensors detect the impending collision, but the distance is too short for emergency braking to prevent a crash. The vehicle has only two options: swerve into oncoming traffic, potentially endangering the lives of drivers, or swerve onto the sidewalk, potentially colliding with the children.
This scenario highlights the uncertainty that autonomous vehicles face in making ethical decisions. The vehicle possesses limited information about the children's age, their awareness of the oncoming vehicle, and the likelihood of them stopping or changing direction. These factors significantly impact the moral implications of each decision, making it challenging to determine the course of action that minimizes harm.
To address this uncertainty, I propose an approach that combines technological advancements with human-centric ethical principles. Firstly, the vehicle should employ advanced sensor technologies and predictive algorithms to gather as much information as possible about the situation – the children's ages, their movements, the speed and direction of oncoming traffic, and any potential hazards on the sidewalk. This comprehensive information will provide a more

nuanced understanding of the potential risks and benefits of each decision.
Secondly, the vehicle's ethical programming should prioritize the preservation of human life, especially the lives of the most vulnerable – in this case, the children. This principle aligns with the fundamental human value of protecting the innocent and upholding the sanctity of human life. While the decision to swerve onto the sidewalk may increase the risk of harming others, it is important to consider the potential harm to the children if the vehicle remains on its current course.
In the face of uncertainty, the vehicle should err on the side of caution, prioritizing the preservation of life whenever possible. This approach aligns with the ethical principle of non-maleficence, which dictates that one should not inflict harm on others. By prioritizing the children's wellbeing, the vehicle demonstrates a commitment to human values while acknowledging the inherent limitations of its decision-making processes.
To effectively implement this approach, programmers should develop a complex algorithm that weighs the potential harm to all parties involved, considering the vehicle's sensors' data, the ethical principles, and the evolving circumstances. This algorithm should continuously evaluate the information as it becomes available and adjust the decision-making process accordingly.
While these technological advancements will undoubtedly enhance the vehicle's ability to make informed decisions, it is essential to acknowledge that complete certainty will remain elusive. In such cases, the vehicle should defer to human judgment. By incorporating a human-in-the-loop system, where an operator can override the vehicle's decisions in life-or-death situations, we can ensure that human values and ethical considerations remain at the forefront of autonomous vehicle operations.